# Generative AI: LLM

What is an LLM ?

It is an AI model that has learned patterns from huge amount of text and can use those patterns to understand and generate new content.




bias
weights
Query
key
Value

Large Language Model

Trained on large amount of internet data and has Large no.of parameters

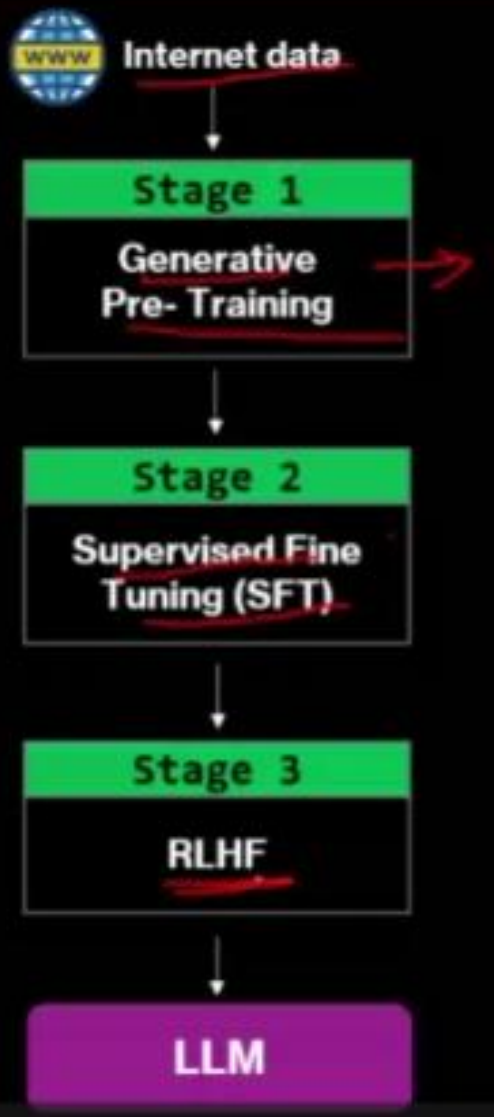
Primarily works with Human language like English, Hindi, French etc.

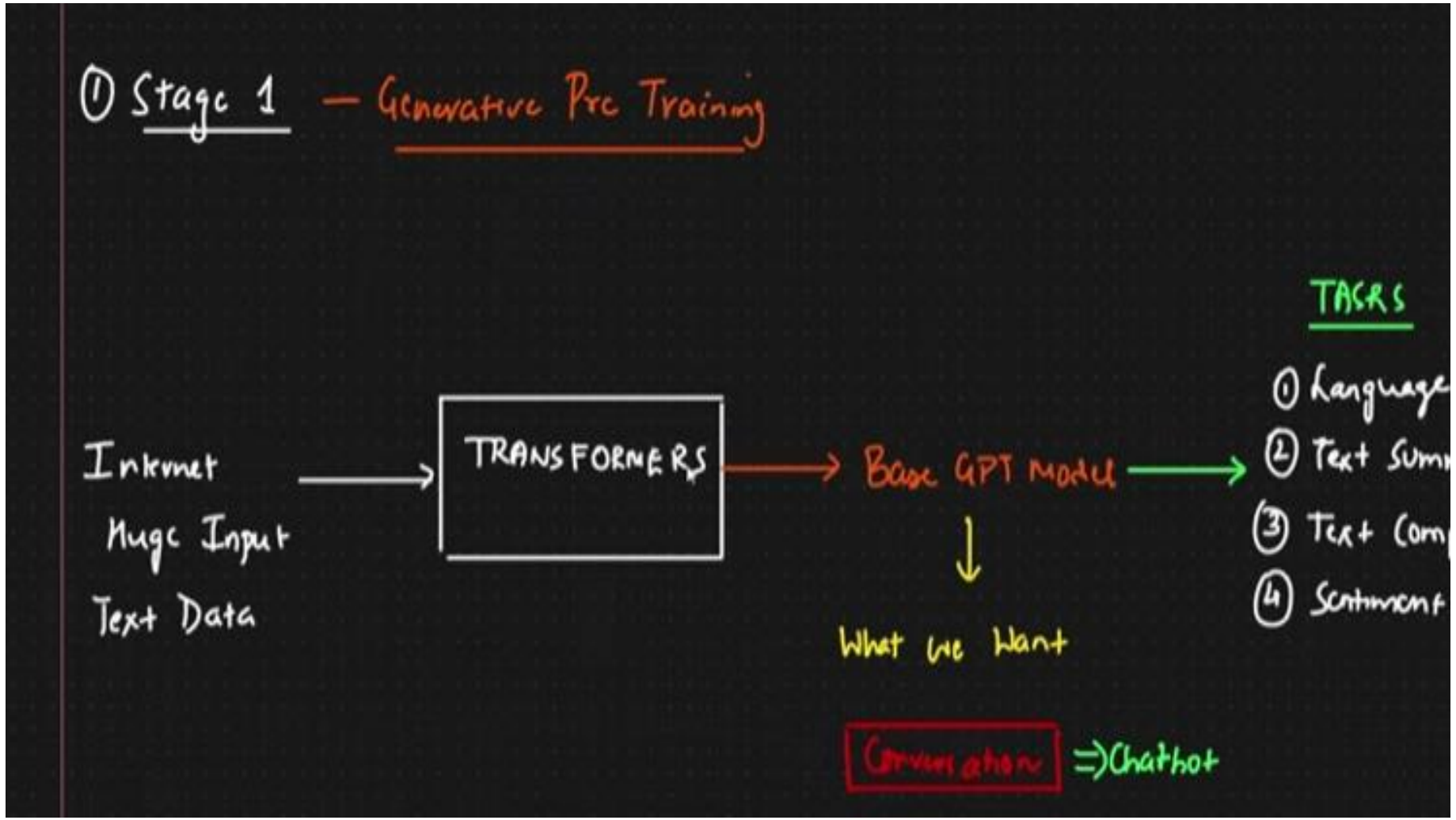
A mathematical model That learns patterns from the data.

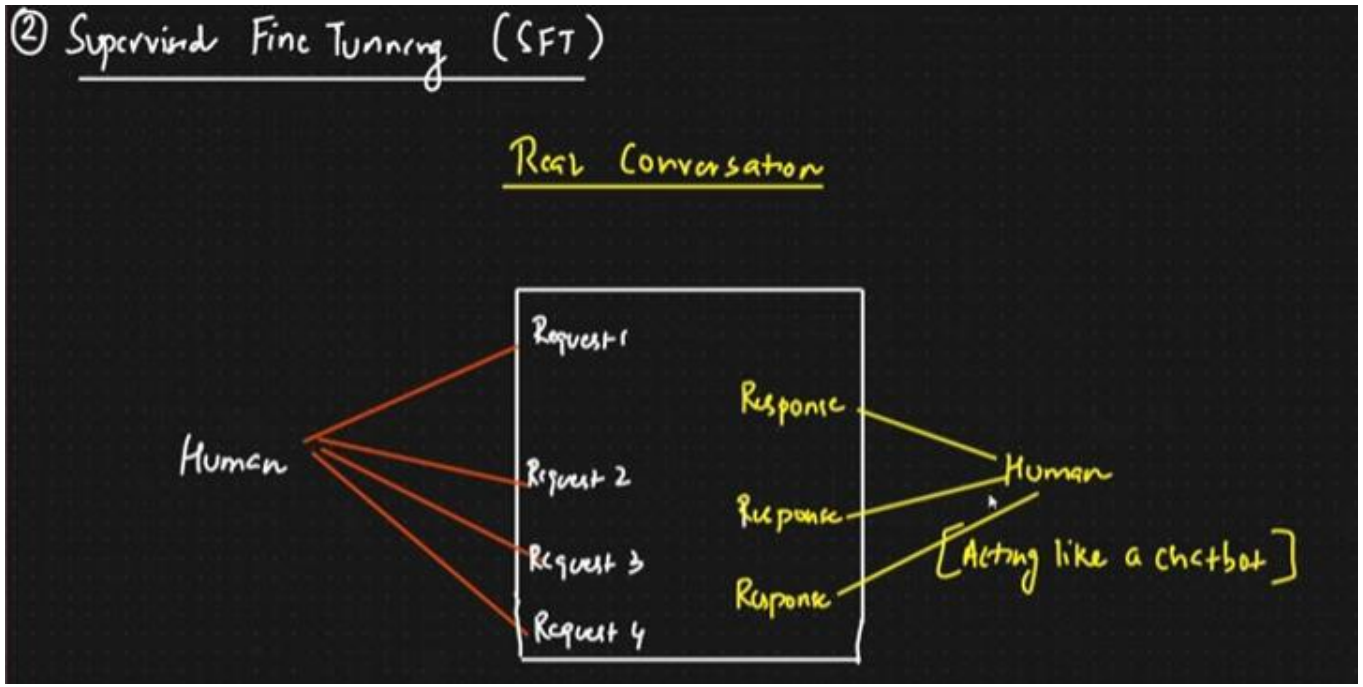
# Generative AI: LLM

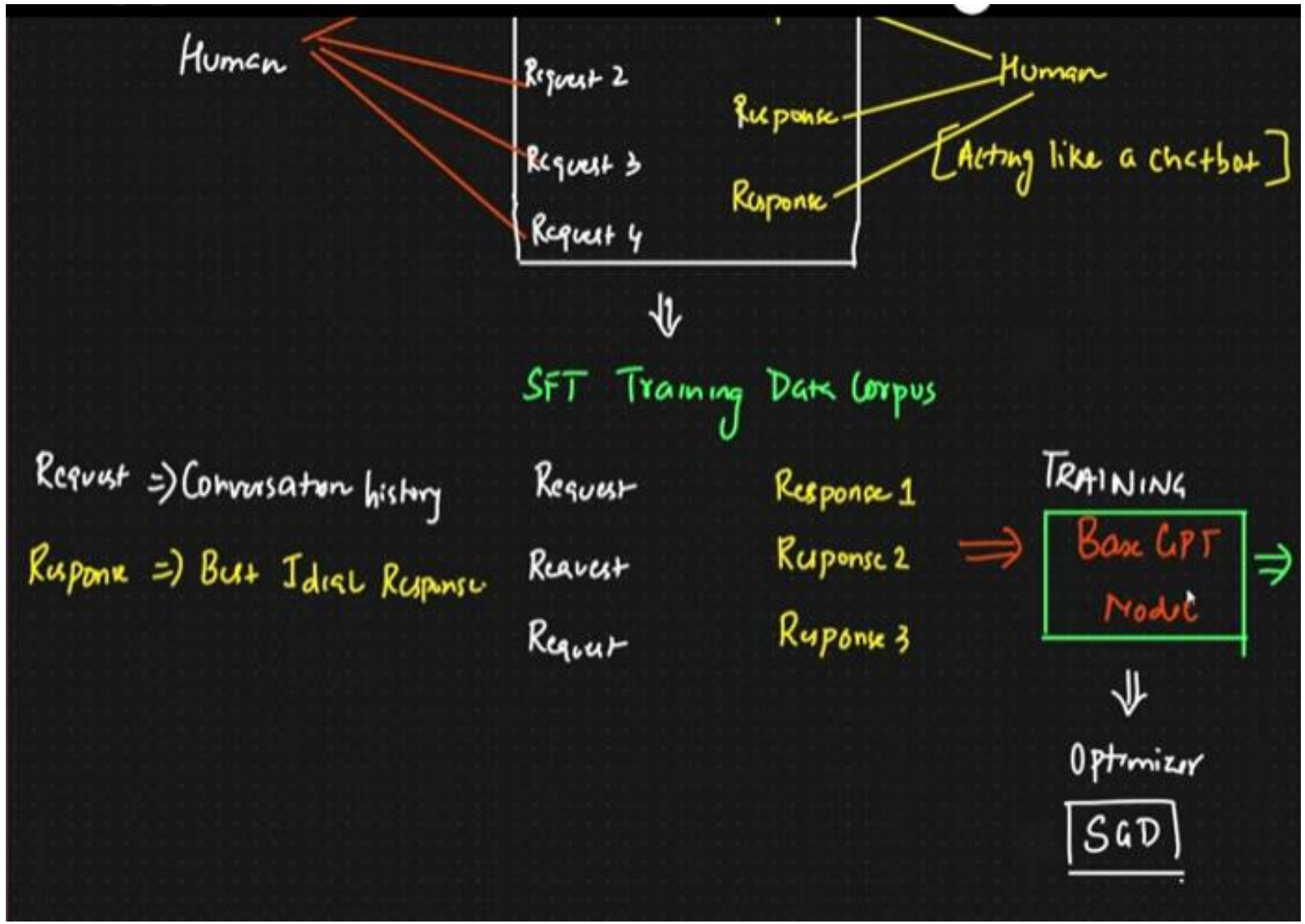
How an LLM is trained?

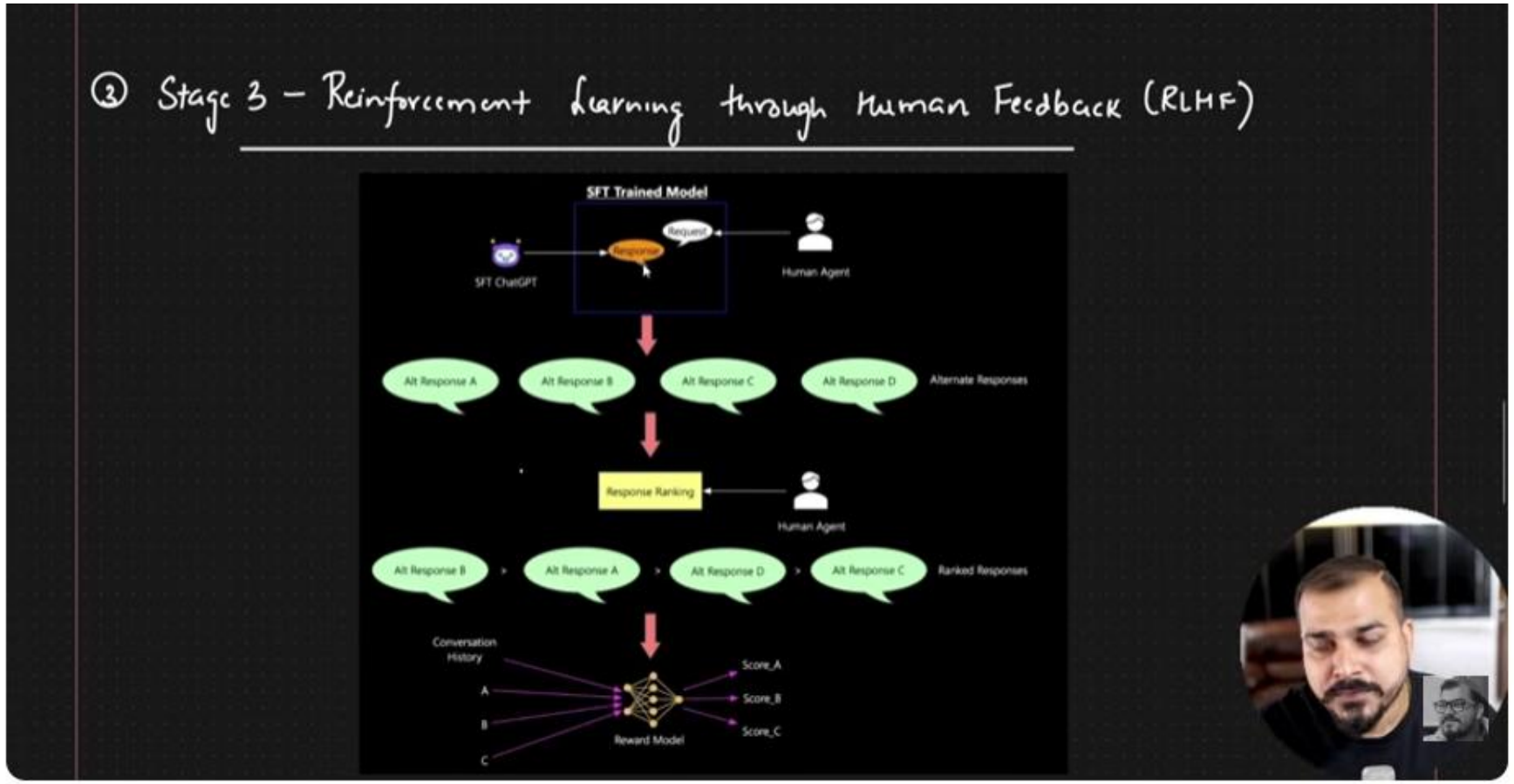



① Stage 1 — Generative Pre Training
Internet
Huge Input
Text Data
TRANSFORMERS
Base GPT Model
What we Want
TASKS
① Language
② Text Summ
③ Text Com
④ Sentiment
Conversation
⇒Chatbot


② Supervised Fine Tunning (SFT)
Real Conversation
Human
Request 1
Request 2
Request 3
Request 4
Response
Response
Response
Human
[Acting like a chatbot]


Human
Request 2
Request 3
Request 4
Response
Response
Human
[Acting like a chatbot]
SFT Training Data Corpus
Request => Conversation history
Response => Best Ideal Response
Request
Request
Request
Response 1
Response 2
Response 3
TRAINING
Base GPT Model
Optimizer
SGD




## How Large Language Models (LLM) In Generative AI Are Trained ?





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What is LLM ? Chatgpt, Gemini, Claude | How it ...

OnlineStudy4u

▷ 13K 3w ago




How LLMs Works? - Overview

Piyush Garg

▷ 564K 1y ago




How LLM Works? Simplified Self Attention | Deep ...

Gate Smashers

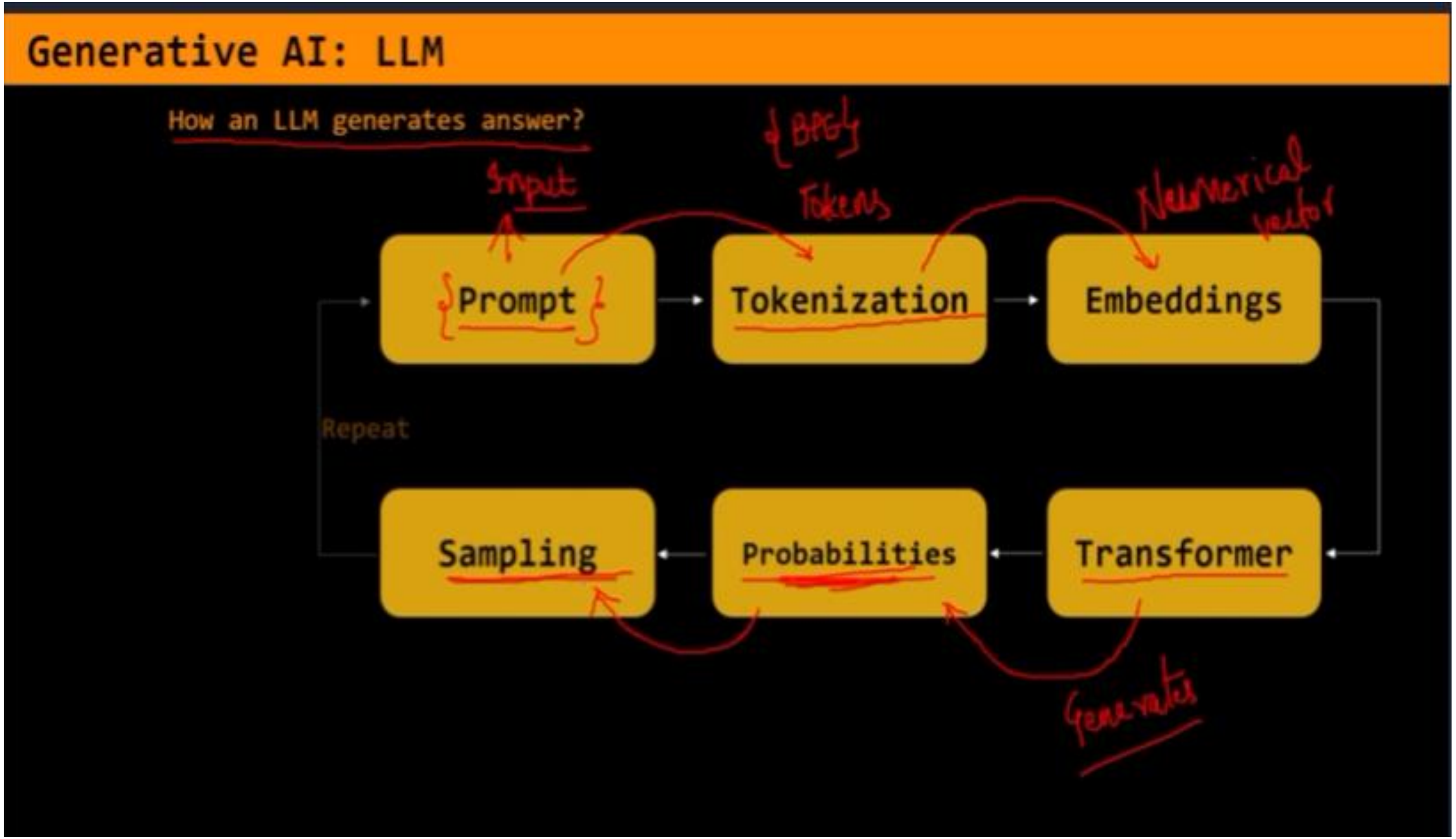

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ENG IN

09:56 AM 06-10-2026

# Generative AI: LLM



# Generative AI: LLM

How an LLM generates answer?

**Prompt**

What is the capital of India?

LLM cannot perform mathematical operations on these English characters, so tokenization should be done

What is the capital of India?

# Generative AI: LLM

How an LLM generates answer?

## Tokenization

What is the capital of India?

↓ tokens

[ “what”, “is”, “the”, “capital”, “of”, “India”, “?” ]

↓ token ID

[ 125, 48, 16, 8291, 72, 5432, 30]

Byte Pair Encoding ( BPE )

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# Generative AI: LLM

How an LLM generates answer?

**Tokenization**

What is the capital of India?

↓ tokens

[ "what", "is", "the", "capital", "of", "India", "?" ]

↓ token ID

[ 125, 48, 16, 8291, 72, 5432, 30]

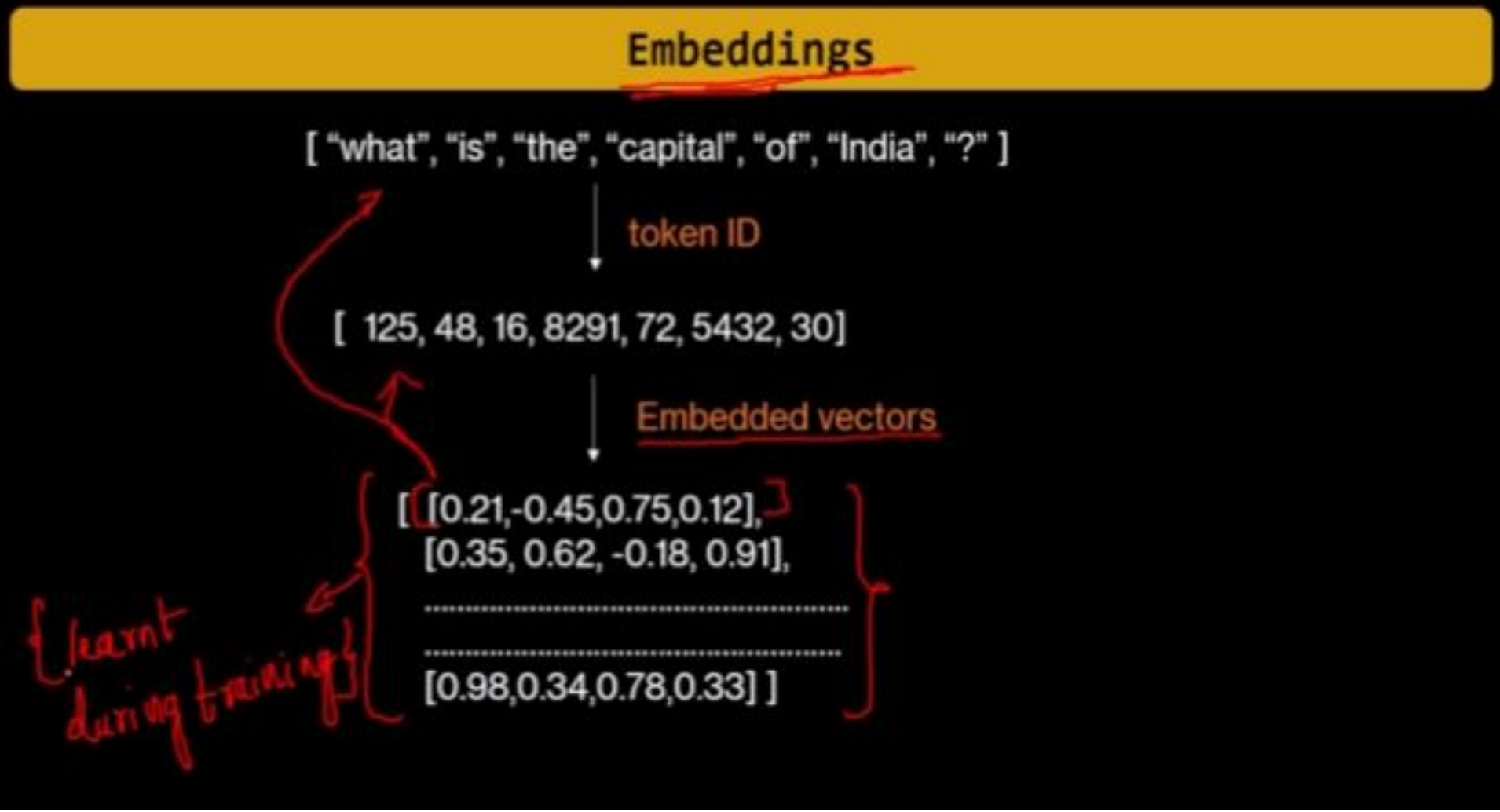
Byte Pair Encoding ( BPE )

These ID's does not contain enough useful information

No meaning

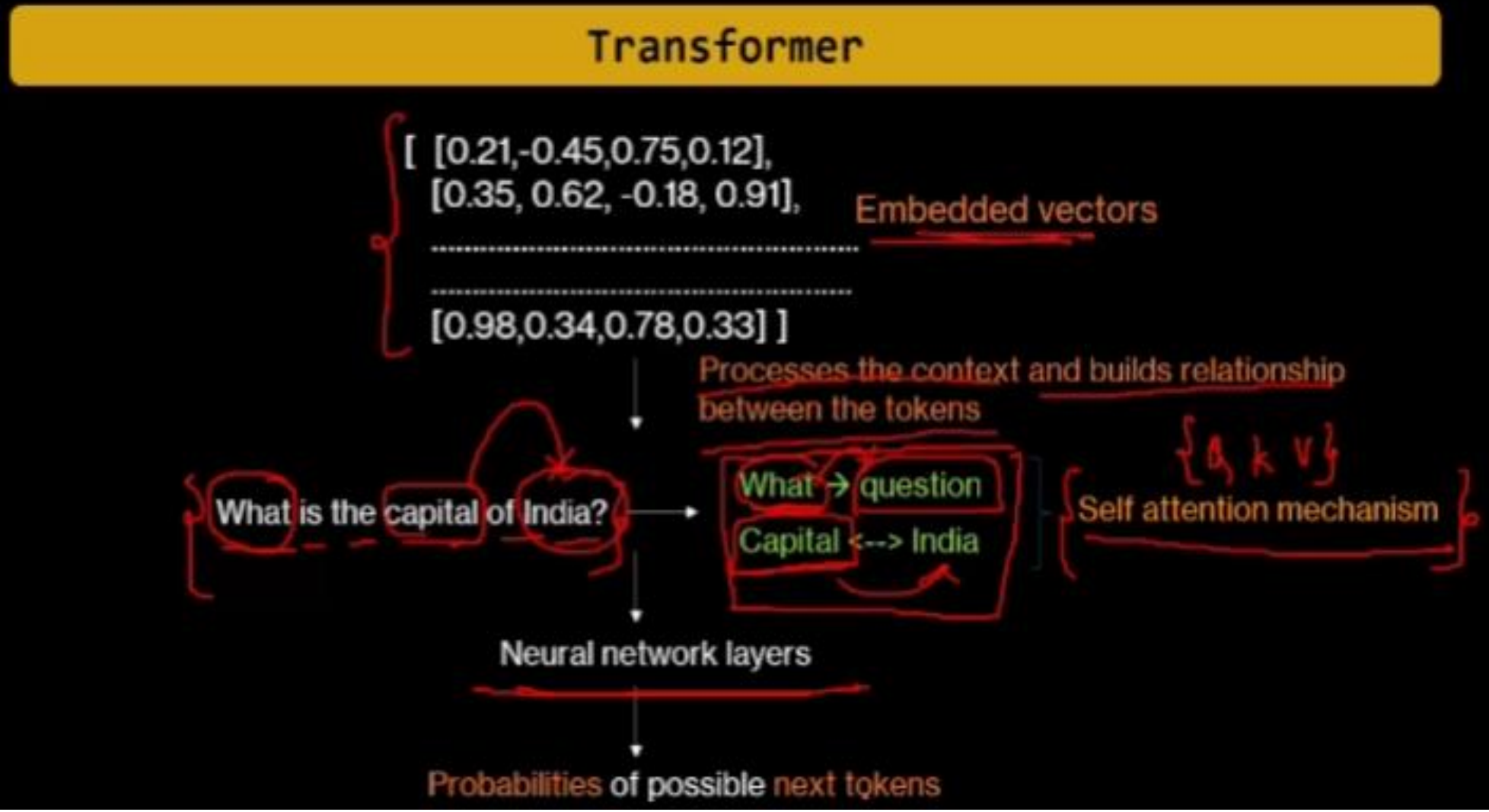
# Generative AI: LLM

How an LLM generates answer?



# Generative AI: LLM

How an LLM generates answer?

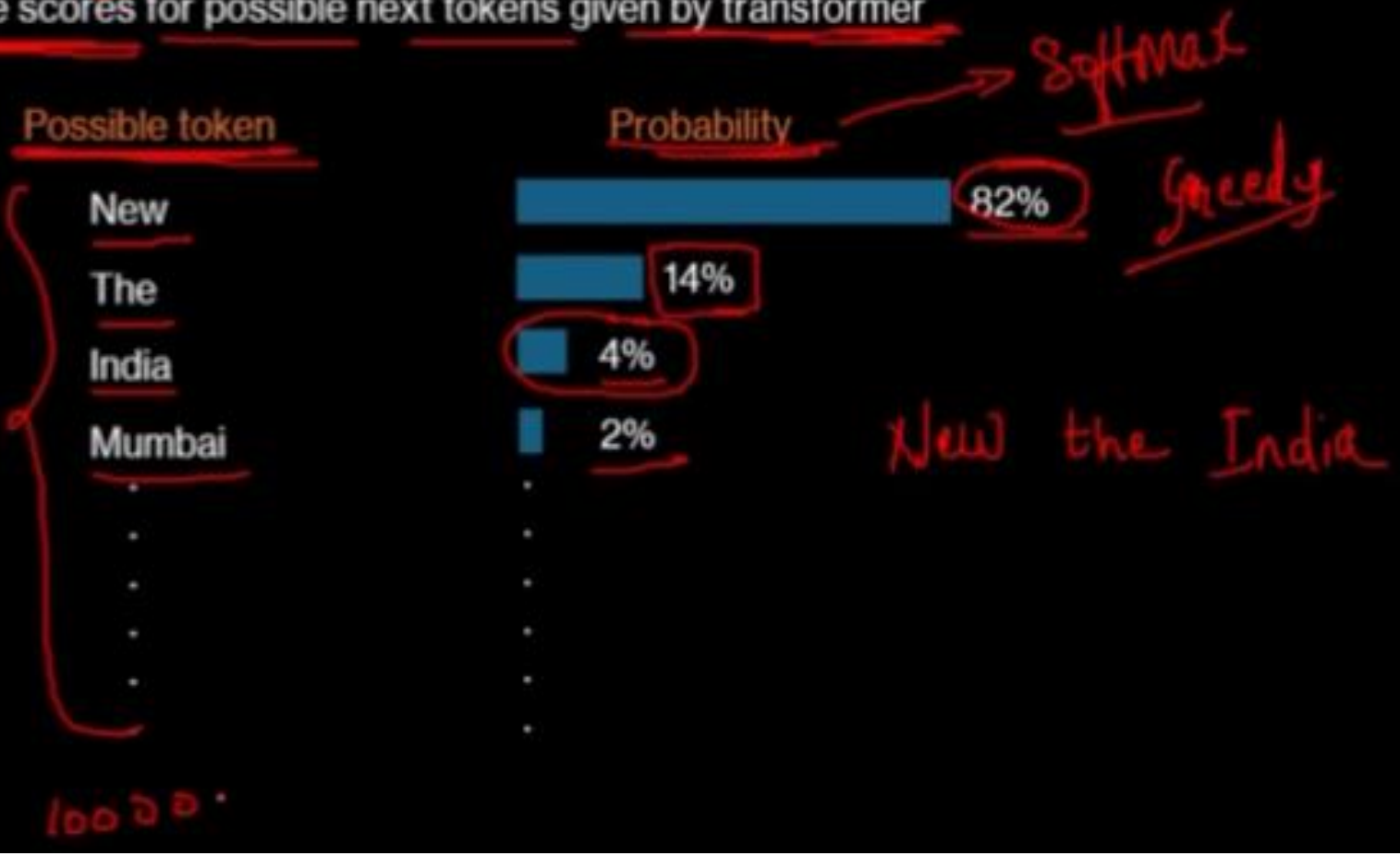


# Generative AI: LLM

How an LLM generates answer?

**Probabilities**

These are scores for possible next tokens given by transformer

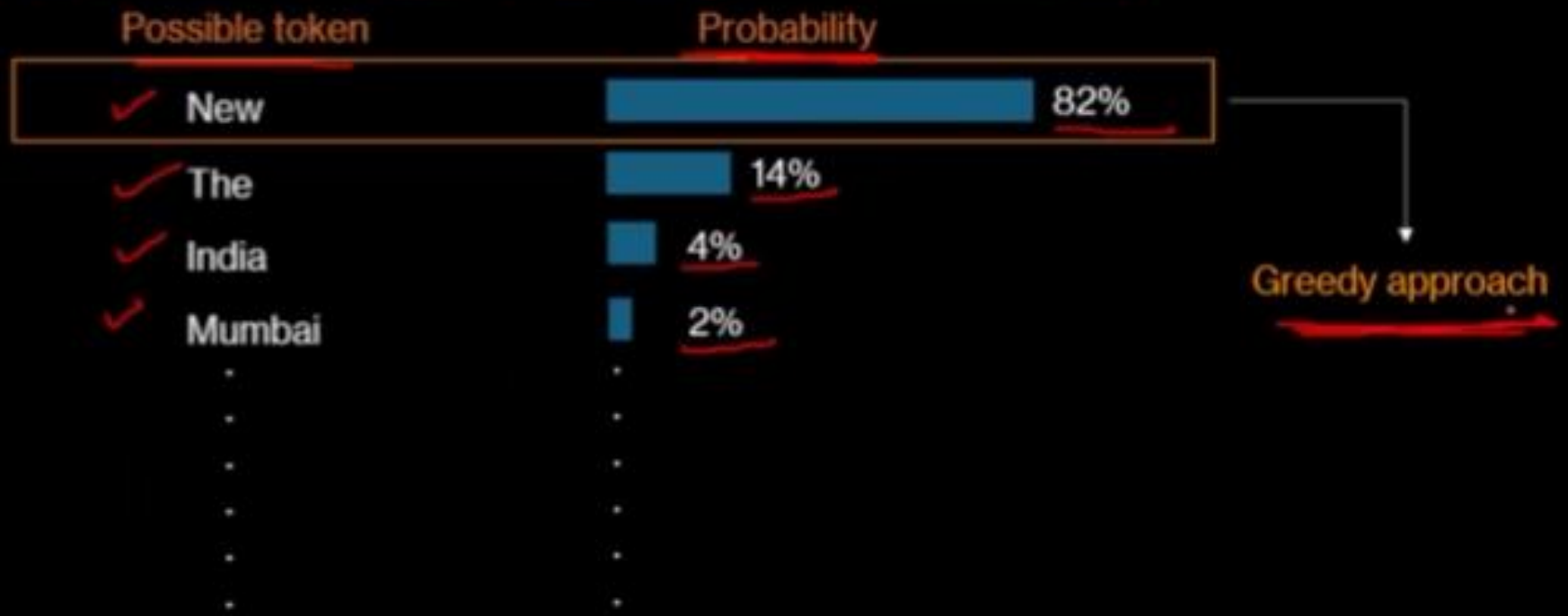


# Generative AI: LLM

How an LLM generates answer?

**Sampling**

To choose one token from these probabilities sampling should be done

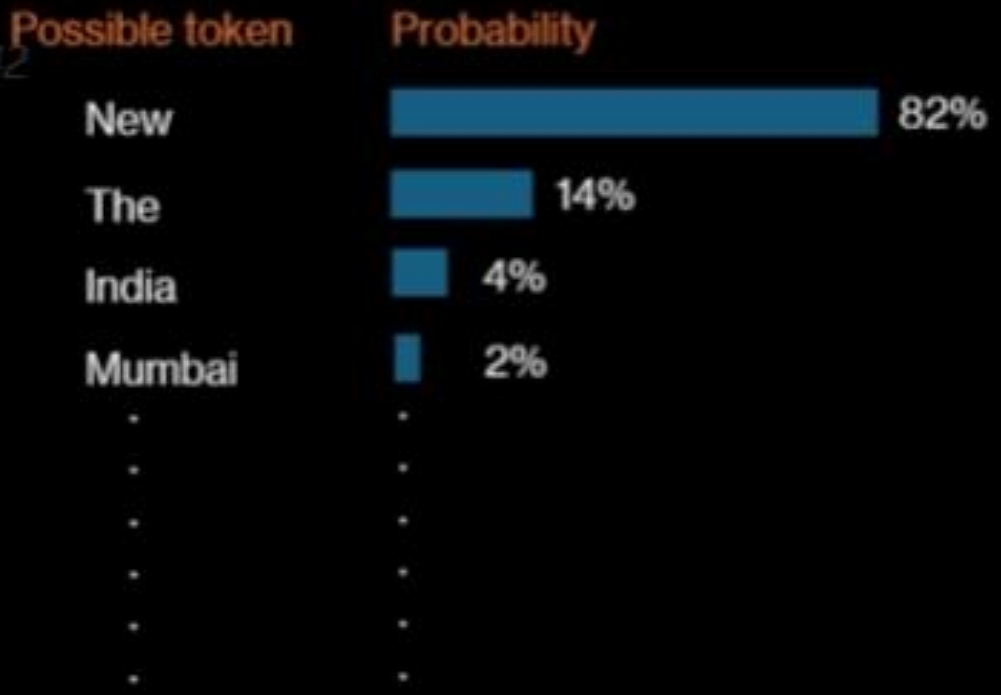


# Generative AI: LLM

How an LLM generates answer?

**Sampling**

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Techniques:

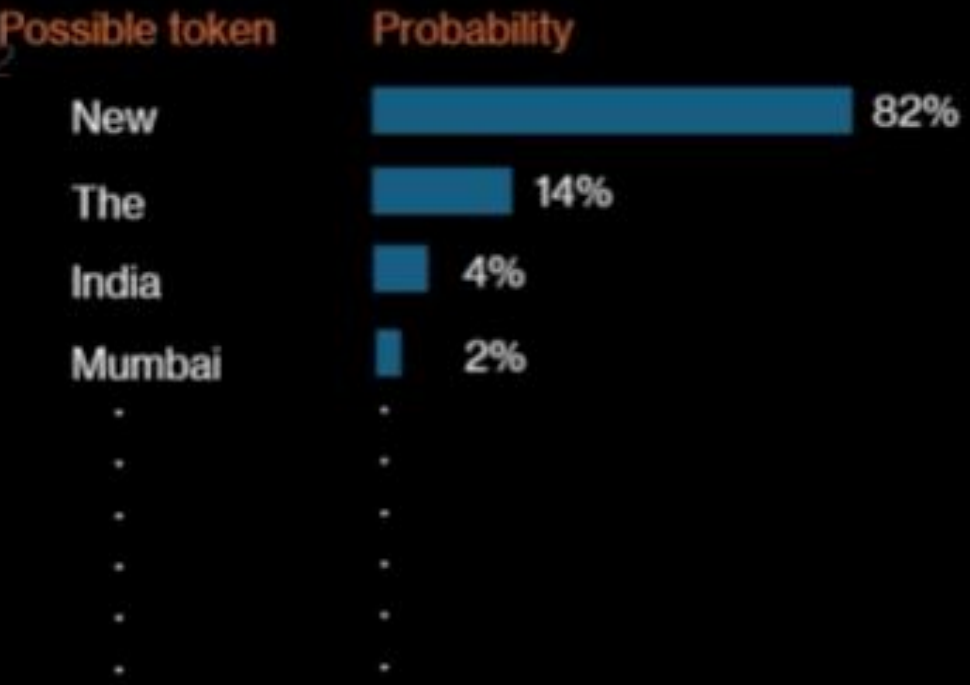
- Temperature
- Top – K
- Top - P

# Generative AI: LLM

How an LLM generates answer?

**Sampling**

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Techniques:

- Temperature
- Top – K
- Top - P

# Generative AI: LLM

How an LLM generates answer?

[0-1]

**Sampling**

| Possible token | Probability |
|---|---|
| New | 82% |
| The | 14% |
| India | 4% |
| Mumbai | 2% |

Temperature: (Hyper-Parameter)

It adjusts how confident the probability distribution is.

Low temperature → chooses high probability tokens

High temperature → chooses low probability tokens

Accurate Answers

Short Poem

0-0.3 (low)
0.4-0.7 (Medium)
0.7-1 (high)

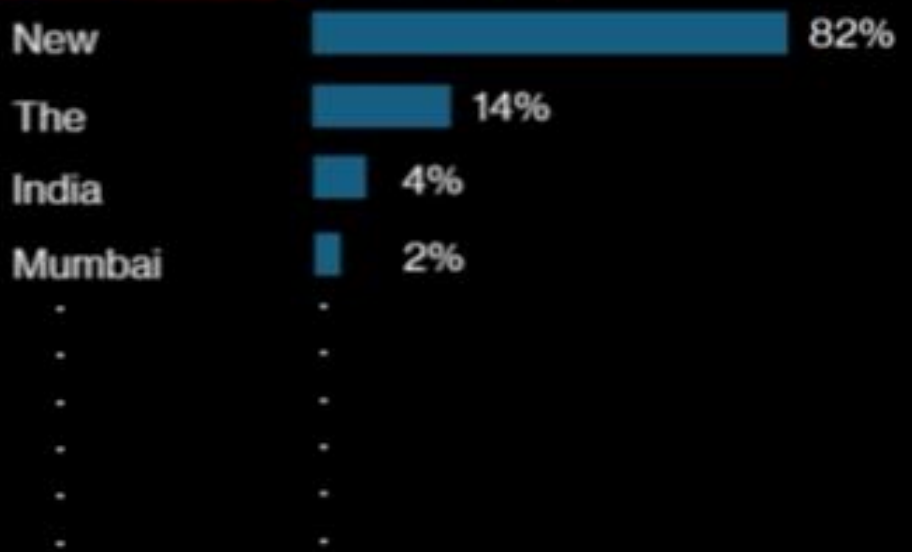
# Generative AI: LLM

How an LLM generates answer?

## Sampling

Top - K:

Model → 50,000 possible tokens




Top – K = 3

# Generative AI: LLM

How an LLM generates answer?

## Sampling

Top - P: It selects smallest group of tokens whose combined probability is closer to top –p value

